1 **(a)** 21 **[B2]**

M1 for 13 or 8

(b) 80 **[B2]**

M1 for 16 or 5

[Total Marks: 4]

2 **(a)** 14 **[B1]**

(b) 59 **[B2]**

If 0 scored, M1 for 64 or 5 seen **[M1]**

(c) 5 **[B2]**

If 0 scored, M1 for $169 - 144$ or 25 seen **[M1]**

[Total Marks: 5]

3 **(a)** $3x + 4 = 64$ **[M1]**

$3x = 60$ **[M1]**

20 **[A1]**

(b) $x^2 = 64 + 36 = 100$ **[M1]**

$10, -10$ **[A1]**

(c) $3\sqrt{x} = 21$ or $\sqrt{x} = 7$ **[M1]**

49 **[A1]**

[Total Marks: 7]

4 **(a)** 125 **[B2]**

If 0 scored, M1 for evaluating $16 - 9 - 2$ to get 5 **[M1]**

(b) -2 **[B3]**

M1 for $\sqrt{64}$ or 8 **[M1]**

M1 for $5 - 9$ or -4 **[M1]**

(c) 729 **[B3]**

M1 for $2\sqrt[3]{y} = 6$ or $\sqrt[3]{y} = 3$ **[M1]**

M1 for $y = 27$ **[M1]**

[Total Marks: 8]

5 **(a)** 5.3990299... **[B2]**

M1 for numerator evaluated to 163.91... or denominator evaluated to 30.359...

(b) 5.399 **[B1]**

(c) 6 **[B2]**

M1 for evaluating $6^4 = 1296$ and $7^4 = 2401$

[Total Marks: 5]

6 **(a)** **(i)** 1.49315... **[B2]**

If 0 scored, M1 for 3.329 seen or 72.25 and 68.921 **[M1]**

(ii) 1.49 **[B1]**

(b) 96 **[B2]**

If 0 scored, M1 for 23.04 or 0.24 seen **[M1]**

[Total Marks: 5]